

Technical & Product Documentation

KDIA Re Park - Customer & Admin Portals

Version: 2.0

Status: Enhanced MVP / Pilot Ready

Last Updated: 09-01-2026

1. Introduction

KDIA Re Park is an energy infrastructure platform that enables consumers to participate in renewable energy generation and allocation models without directly owning physical assets.

The platform is designed to:

- Provide transparent access to energy allocation and consumption data
- Enable customer participation in clean energy programs
- Support administrative operations for KDIA Re Park teams
- Maintain clear separation between informational energy data and actual electricity billing, which continues to be handled by the respective DISCOMs

2. Platform Components

The KDIA Re Park system consists of two primary portals:

2.1 Customer Portal

A secure interface for consumers to:

- Register and enroll into KDIA Re Park infrastructure programs
- View allocated energy units and consumptions insights
- Access infrastructure reports
- Raise and track support requests

2.2 Admin Portal

An internal operational interface for KDIA administrator to:

- Manage customers and their profiles
- Handle customer support tickets

- Manage allocations and consumption records
- Maintain auditability and traceability

3. System Architecture

Architecture Pattern

Client - Server Architecture

Frontend

- React (Vite)
- Tailwind CSS
- Recharts (charts & data visualization)

Backend

- [Node.js](#)
- [Express.js](#)
- JWT-based authentication
- Role based access control (Customer / Admin)

Database

- SQLite (MVP / Pilot Phase)
- Designed to be portable to enterprise RDBMS later

Security Measures

- Password hashing using bcrypt
- Protected API routes
- Input validation at UI and API layers
- Admin action logging

4. User roles and permissions

Customer Role

Customers can:

- Register and authenticate
- View personal dashboard
- View energy allocation and consumption data
- Download infrastructure reports

- Raise support tickets
- Track ticket status
- Revoke unresolved tickets

Admin Role

Admins can:

- Authenticate via Admin Portal
- View and manage all customers
- Edit customer personal and address details
- Reset or change customer passwords
- View and respond to support tickets
- Categorize and resolve tickets
- Access audit logs

5. Customer Registration (Signup / Infrastructure Enrollment)

5.1 Personal Details

- Full name (mandatory)
- Date of birth (mandatory, must be valid and not future dated)
- Gender (mandatory)

5.2 Contact Details

- Email address (mandatory, validated format)
- Mobile number (mandatory)
 - Numeric only
 - Exactly 10 digits
 - Inline error messaging

5.3 Address & Service Eligibility Details

(Required to provide KDIA Re Park services)

- Address Line 1 (mandatory)
- Address Line 2 (optional)
- City (mandatory)
- State (mandatory)
- Pin Code (mandatory, 6 digits)
- Location type (residential / commercial)

These details are used for service eligibility, compliance and operational planning.

5.4 Security Credentials

- Password (mandatory)
- Requirements:
 - Minimum 8 characters
 - Uppercase + lowercase letters
 - At least one number
 - At least one special character

Validation Rules

- Signup is blocked until all mandatory fields are valid
- Inline error messages are shown
- Logical field order:
 - a. Personal details
 - b. Contact details
 - c. Address & service details
 - d. Password

6. Customer Login & Account Recovery

Login

- Email or Mobile Number
- Password

Forgot Password

- Availability on login screen
- Opens Account Recovery modal
- Displays:
 - KDIA support email
 - KDIA support phone number
- No redirection to support page
- Password recovery handled manually by KDIA team

7. Customer Dashboard

Core Features

- Allocated energy overview
- Consumption summary
- Infrastructure ledger / history

- Downloadable reports

Action cards

- Infrastructure report
- Increase allocation
- Support & Help

All cards:

- Follow identical hover behaviour
- Use unified green branding
- Use consistent icons
- No black or inconsistent hover states.

8. Support & Help - Customer Portal

Raise Support Requests

Customers can raise tickets categorized as:

- General question
- Consumption clarification
- Allocation related
- other

Ticket Directory (Customer)

Customers can view:

- Ticket ID
- Category
- Status ({Pending / In Progress / Resolved / Revoked})
- Last updated timestamp
- Admin replies (if any)

Ticket Actions

- Revoke ticket (only if resolved)

9. Support and Ticket Management - Admin Portal

Ticket Listing

Admins can:

- View all customer tickets

- Filter by category and status
- Sort by date

Ticket details

Admins can:

- View full ticket context
- Reply to customer tickets
- Update ticket status

Ticket categories

- General question
- Consumption clarification
- Allocation related
- Other

10. Admin Portal - Customer Management

Admins can:

- View customer profiles
- Edit customer personal details
- Update address information
- Reset or change customer passwords

Security:

- Passwords are never displayed
- Sensitive actions require confirmation
- All actions are logged

11. Admin Authentication & UI

Admin Login

- Secure credentials
- Green primary CTA
- professional , neutral UI

UI Design Principles

- No white containers behind logos
- Unified color system
- Clean sidebar design
- Calm, data-focused layout

12. UI & UX Philosophy

- Infrastructure-first
- Neutral, professional tone
- Minimal visual noise
- Clear hierarchy and readability
- Responsive across devices.

13. Data Disclaimer

All values displayed in the KDIA Re Park portal are indicative and informational only.

Electricity billing, tariffs, and collections continue to be handled by the respective DISCOMs.

14. Audit and Traceability

- Admin actions are logged
- Ticket replies are traceable
- Customer data changes are auditable

15. Conclusion

The KDIA Re Park platform has evolved into a robust, pilot ready infrastructure system that balances customer transparency with administrative control.

It is suitable for:

- Internal demos
- Pilot deployments
- Stakeholder reviews
- Operational onboarding